

A positive-conservative Fourier optimization method for ground states of Bose–Einstein condensates with higher-order interactions

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Abstract

We develop a positive-conservative Fourier optimization (PCFO) method for computing ground states of Bose–Einstein condensates with higher-order interactions. In the density formulation, the kinetic term is singular at vacuum. We regularize this term through a positive concave denominator and combine it with a compatible convex regularization of the potential to remove its nonzero vacuum slope. The resulting energies preserve convexity, and their fixed-box Γ -convergence to the original density energy is established as the regularization parameters vanish. The Fourier discretization preserves mass exactly and imposes positivity at the collocation points. To avoid the large curvature introduced by the potential regularization, we place this term together with the constraints in the proximal part of an accelerated first-order method and use a Newton–PCG refinement for high-accuracy stationarity. Numerical experiments demonstrate spectral-type convergence for fixed smooth regularized problems, examine the effects of the two regularization scales, and confirm the efficiency of the two-stage optimization strategy.

Key words. Bose–Einstein condensate; higher-order interaction; density formulation; convex optimization; Fourier pseudospectral method; positivity preserving.

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1 Introduction

Bose–Einstein condensation, first realized experimentally in dilute atomic gases in 1995 [2], provides a fundamental setting for the study of macroscopic quantum phenomena. In the dilute and low-temperature regime, Bose–Einstein condensates are commonly described by the Gross–Pitaevskii equation [5]. With tunable interactions, in particular near Feshbach resonances [15], finite-range corrections to the standard contact interaction may become relevant [19, 16, 27, 25]. These corrections lead to modified Gross–Pitaevskii models with higher-order interactions.

We consider the modified Gross–Pitaevskii model with higher-order interactions (HOI), whose ground-state energy is

$$E(\phi) = \int_{\mathbb{R}^d} \left[\frac{1}{2} |\nabla \phi|^2 + V(x) |\phi|^2 + \frac{\beta}{2} |\phi|^4 + \frac{\delta}{2} |\nabla(|\phi|^2)|^2 \right] dx, \quad (1.1)$$

subject to

$$\|\phi\|_{L^2}^2 = 1. \quad (1.2)$$

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Here $V \geq 0$ is a confining potential, $\beta \in \mathbb{R}$ is the contact-interaction strength, and $\delta > 0$ is the higher-order interaction strength. Ground-state properties and asymptotic regimes for this model have been studied in [22, 6]. In this work, we focus on the repulsive regime $\beta \geq 0$.

A ground state may be chosen real and nonnegative. Introducing the density $\rho = |\phi|^2$ then gives the equivalent energy

$$\mathcal{E}_0(\rho) = \int_{\mathbb{R}^d} \left[\frac{|\nabla \rho|^2}{8\rho} + V(x)\rho + \frac{\beta}{2}\rho^2 + \frac{\delta}{2}|\nabla \rho|^2 \right] dx, \quad (1.3)$$

over

$$\mathcal{A} = \left\{ \rho \in H^1(\mathbb{R}^d) : \rho \geq 0, \quad \int_{\mathbb{R}^d} \rho \, dx = 1, \quad \int_{\mathbb{R}^d} V\rho \, dx < \infty \right\}. \quad (1.4)$$

The singular kinetic term in (1.3) is understood in the extended sense specified in Section 2. For $\beta \geq 0$, the density formulation has a convex structure that is particularly attractive for ground-state computation, but the kinetic term becomes singular at vacuum.

Ground-state computation for the Gross–Pitaevskii equation has been studied extensively. Early approaches include direct energy minimization and normalized gradient-flow methods [10, 8], together with spectrally accurate discretizations [7]. For strongly nonlinear or rotating condensates, Sobolev-gradient methods, preconditioned spectral solvers, nonlinear conjugate-gradient methods, regularized Newton methods, and Riemannian optimization have been developed [17, 3, 4, 26, 18]. More recent work includes normalized Sobolev-gradient flows and nonlinear eigenvalue iterations with convergence analysis [20, 1, 14]. Fourier pseudospectral discretizations combined with manifold optimization have also been applied to more general spinor condensates [23].

For the HOI model, a normalized gradient-flow method with an attractive–repulsive splitting was developed in [21]. W. Bao and X. Ruan [9] subsequently introduced a regularized density formulation, in which the ground-state problem is treated as a convex minimization problem using a shifted kinetic regularization, a second-order finite-difference discretization, and an accelerated projected-gradient method. A related square-root regularization of the potential was also considered there as a numerical enhancement.

The present work builds on this density formulation with the aim of achieving high spatial accuracy and high-accuracy stationarity. Regularizing the kinetic term alone leaves a nonzero potential slope at vacuum, which may produce an active positivity set and limit the regularity seen by a Fourier discretization. Moreover, decreasing the potential-regularization scale introduces large local curvature, while the final stationarity can become difficult to resolve efficiently by a first-order iteration alone.

To address these issues, we consider a general convexity-preserving regularization of the kinetic denominator together with an independently scaled convex regularization of the potential, and establish the fixed-box Γ -convergence of the jointly regularized energies. The resulting problem is discretized by a Fourier pseudospectral method with exact mass conservation and nodal positivity. A potential-proximal accelerated iteration is used for the main minimization, followed by a Newton–PCG refinement for high-accuracy stationarity. Numerical experiments examine spatial accuracy, regularization effects, and solver performance, and illustrate the method for two-dimensional trapping potentials.

The rest of the paper is organized as follows. Section 2 introduces the convex regularizations and establishes their fixed-box variational convergence. Section 3 presents the Fourier pseudospectral discretization and the positive mass-conservative projection. Section 4 develops the potential-proximal FISTA method and the Newton–PCG refinement. Section 5 reports the numerical results, and Section 6 concludes the paper.

2 Density formulation and convex regularization

In this section, we formulate the ground-state problem in terms of the density and study convex regularizations of the singular kinetic term and the potential term. We first describe the convex structure of the density energy and construct the two regularizations. We then establish their variational convergence on a fixed periodic box as the regularization parameters vanish.

2.1 Convex regularization of the kinetic and potential terms

We first regularize the density kinetic term while preserving its convex structure, and then combine it with a compatible regularization of the potential term near vacuum.

Kinetic regularization. We begin by extending the density kinetic term to possible vacuum regions. For $s \geq 0$ and $q \in \mathbb{R}^d$, let

$$F_0(s, q) = \begin{cases} \frac{|q|^2}{8s}, & s > 0, \\ 0, & s = 0, \ q = 0, \\ +\infty, & s = 0, \ q \neq 0. \end{cases} \quad (2.1)$$

With this convention, the density energy in (1.3) can be written as

$$\mathcal{E}_0(\rho) = \int_{\mathbb{R}^d} \left[F_0(\rho, \nabla \rho) + V\rho + \frac{\beta}{2}\rho^2 + \frac{\delta}{2}|\nabla \rho|^2 \right] dx, \quad \rho \in \mathcal{A}. \quad (2.2)$$

We first record the convexity property that motivates the regularization introduced below.

Proposition 2.1 (Convexity of the density energy). *Assume $\beta \geq 0$ and $\delta > 0$. Then $\mathcal{E}_0$ is strictly convex on $\mathcal{A}$ and hence has at most one minimizer.*

Proof. The map $(s, q) \mapsto |q|^2/s$ is the perspective of the convex quadratic function $q \mapsto |q|^2$, and (2.1) is its lower-semicontinuous convex extension to $s = 0$. The potential term is linear, while the remaining terms are convex.

The term $\frac{\delta}{2}\|\nabla \rho\|_{L^2}^2$ provides strict convexity. Indeed, equality in its convexity inequality for two densities in $\mathcal{A}$ implies that their gradients coincide almost everywhere. Their difference is therefore constant. Since both densities are integrable on $\mathbb{R}^d$, this constant must be zero. $\square$

We now regularize the kinetic term in a way that preserves this convexity. For $s \geq 0$ and $q \in \mathbb{R}^d$, set

$$F_\varepsilon(s, q) = \frac{|q|^2}{8r_\varepsilon(s)}, \quad (2.3)$$

where r_ε is a positive regularization of the denominator.

Definition 2.2 (Admissible denominator). A family $\{r_\varepsilon\}_{\varepsilon>0}$ is called admissible if, for every $\varepsilon > 0$,

$$r_\varepsilon \in C^1([0, \infty)), \quad r_\varepsilon > 0, \quad r'_\varepsilon \geq 0, \quad r_\varepsilon \text{ is concave}, \quad (2.4)$$

and, for every $s \geq 0$,

$$s \leq r_{\varepsilon_1}(s) \leq r_{\varepsilon_2}(s) \quad \text{for } 0 < \varepsilon_1 \leq \varepsilon_2, \quad r_\varepsilon(s) \rightarrow s \quad \text{as } \varepsilon \rightarrow 0^+. \quad (2.5)$$

The concavity of the denominator is precisely what preserves the convexity of the regularized kinetic term.

Lemma 2.3 (Convexity of the regularized kinetic integrand). *Let r_ε be admissible in the sense of Definition 2.2. Then*

$$F_\varepsilon(s, q) = \frac{|q|^2}{8r_\varepsilon(s)}$$

is jointly convex in $(s, q) \in [0, \infty) \times \mathbb{R}^d$.

Proof. Let $0 \leq \theta \leq 1$ and set

$$s_\theta = \theta s_1 + (1 - \theta)s_2, \quad q_\theta = \theta q_1 + (1 - \theta)q_2.$$

By the concavity of r_ε ,

$$r_\varepsilon(s_\theta) \geq \theta r_\varepsilon(s_1) + (1 - \theta)r_\varepsilon(s_2).$$

Hence

$$F_\varepsilon(s_\theta, q_\theta) \leq \frac{|q_\theta|^2}{8[\theta r_\varepsilon(s_1) + (1 - \theta)r_\varepsilon(s_2)]}.$$

The weighted Cauchy–Schwarz inequality gives

$$\frac{|q_\theta|^2}{\theta r_\varepsilon(s_1) + (1 - \theta)r_\varepsilon(s_2)} \leq \theta \frac{|q_1|^2}{r_\varepsilon(s_1)} + (1 - \theta) \frac{|q_2|^2}{r_\varepsilon(s_2)},$$

which proves the result. $\square$

The simplest example is the shifted denominator introduced in [9],

$$r_\varepsilon^{\text{sh}}(s) = s + \varepsilon. \tag{2.6}$$

For comparison, we also consider the family

$$r_{\varepsilon, m}(s) = \begin{cases} s + \varepsilon - \frac{\varepsilon}{m+1} \left(1 - \frac{s}{\varepsilon}\right)^{m+1}, & 0 \leq s \leq \varepsilon, \\ s + \varepsilon, & s > \varepsilon, \end{cases} \quad m = 1, 2. \tag{2.7}$$

These functions belong to $C^m([0, \infty))$ and satisfy Definition 2.2. The cases $m = 1, 2$ are included in the numerical comparison below.

Potential regularization. The kinetic regularization removes the singularity at $\rho = 0$, but the linear potential term $V\rho$ still has the nonzero vacuum derivative V . In the discrete problem this may produce an active positivity set and thereby limit the attainable Fourier accuracy. We therefore regularize the potential contribution near vacuum by replacing ρ with

$$p_\sigma(s) = \sqrt{s^2 + \sigma^2} - \sigma, \quad s \geq 0. \tag{2.8}$$

A square-root smoothing of this type was already considered as a numerical enhancement in [9]. Here σ is treated as an independent regularization scale and the two regularizations are analyzed jointly. It satisfies

$$p_\sigma(0) = p'_\sigma(0) = 0, \quad p''_\sigma(s) = \frac{\sigma^2}{(s^2 + \sigma^2)^{3/2}} \geq 0. \tag{2.9}$$

Hence Vp_σ is convex whenever $V \geq 0$. Moreover, $p_\sigma(s) \rightarrow s$ as $\sigma \rightarrow 0^+$. For the convergence analysis below, it is sufficient to assume that $\sigma = \sigma(\varepsilon) \rightarrow 0$ as $\varepsilon \rightarrow 0^+$.

Combining the two regularizations, we obtain

$$\mathcal{E}_{\varepsilon,\sigma}(\rho) = \int_{\mathbb{R}^d} \left[F_\varepsilon(\rho, \nabla \rho) + V(x)p_\sigma(\rho) + \frac{\beta}{2}\rho^2 + \frac{\delta}{2}|\nabla \rho|^2 \right] dx, \quad \rho \in \mathcal{A}. \quad (2.10)$$

The convex structure of the original density problem is retained by the regularized energy.

Proposition 2.4 (Convexity of the regularized energy). *Assume that r_ε is admissible in the sense of Definition 2.2, $V \geq 0$, $\beta \geq 0$, $\delta > 0$, and $\sigma > 0$. Then $\mathcal{E}_{\varepsilon,\sigma}$ is strictly convex on $\mathcal{A}$. If $\beta > 0$, it is strongly convex with respect to the L^2 norm.*

Proof. By Lemma 2.3, the kinetic contribution is convex. The potential term is convex because p_σ is convex and $V \geq 0$, while the remaining terms are convex. Strict convexity follows from the δ term as in Proposition 2.1. If $\beta > 0$, the term $\frac{\beta}{2}\|\rho\|_{L^2}^2$ gives strong convexity in L^2 . $\square$

The two properties of p_σ play different roles. The condition $p'_\sigma(0) = 0$ removes the nonzero vacuum slope of the potential term, while the curvature $p''_\sigma(0) = 1/\sigma$ motivates the proximal treatment introduced in Section 4.1.

2.2 Fixed-box Γ -convergence

We next study the limit of the regularized problem as the regularization parameters vanish. Since the Fourier discretization considered below is posed on a periodic computational box, we carry out the variational analysis for a fixed $D_L = [-L, L]^d$. This separates the regularization limit from the choice of computational box. We use the standard notion of Γ -convergence in the strong $L^2(D_L)$ topology; see, e.g., [12].

Define

$$\mathcal{A}_L = \left\{ \rho \in H_{\text{per}}^1(D_L) : \rho \geq 0 \text{ a.e., } \int_{D_L} \rho \, dx = 1 \right\}. \quad (2.11)$$

The corresponding fixed-box energies are

$$\begin{aligned} \mathcal{E}_{0,L}(\rho) &= \int_{D_L} \left[F_0(\rho, \nabla \rho) + V\rho + \frac{\beta}{2}\rho^2 + \frac{\delta}{2}|\nabla \rho|^2 \right] dx, \\ \mathcal{E}_{\varepsilon,\sigma,L}(\rho) &= \int_{D_L} \left[F_\varepsilon(\rho, \nabla \rho) + Vp_\sigma(\rho) + \frac{\beta}{2}\rho^2 + \frac{\delta}{2}|\nabla \rho|^2 \right] dx, \quad \rho \in \mathcal{A}_L. \end{aligned} \quad (2.12)$$

The potential regularization introduces only a uniformly small perturbation on a fixed box. Indeed, for every $s \geq 0$,

$$0 \leq s - p_\sigma(s) \leq \sigma, \quad \left| \int_{D_L} V[p_\sigma(\rho) - \rho] \, dx \right| \leq \sigma \int_{D_L} V \, dx. \quad (2.13)$$

This estimate allows the kinetic and potential regularizations to be treated separately in the convergence argument.

Theorem 2.5 (Γ -convergence of the joint regularization). *Let $V \in C(D_L)$ with $V \geq 0$, $\beta \geq 0$, and $\delta > 0$. Let $\{r_\varepsilon\}$ be admissible in the sense of Definition 2.2, and let $\sigma(\varepsilon) > 0$ satisfy $\sigma(\varepsilon) \rightarrow 0$ as $\varepsilon \rightarrow 0^+$. Extend $\mathcal{E}_{\varepsilon,\sigma(\varepsilon),L}$ and $\mathcal{E}_{0,L}$ to $L^2(D_L)$ by setting them equal to $+\infty$ outside $\mathcal{A}_L$. Then*

$$\mathcal{E}_{\varepsilon,\sigma(\varepsilon),L} \xrightarrow{\Gamma} \mathcal{E}_{0,L} \quad \text{in the strong } L^2(D_L) \text{ topology.}$$

Moreover, the family $\{\mathcal{E}_{\varepsilon,\sigma(\varepsilon),L}\}_{\varepsilon>0}$ is equicoercive in $L^2(D_L)$.

Proof. We first consider the kinetic regularization alone and keep the potential term unchanged. Define

$$\tilde{\mathcal{E}}_{\varepsilon,L}(\rho) = \int_{D_L} \left[F_{\varepsilon}(\rho, \nabla \rho) + V\rho + \frac{\beta}{2}\rho^2 + \frac{\delta}{2}|\nabla \rho|^2 \right] dx.$$

We first prove the liminf inequality. Let $\varepsilon_n \rightarrow 0$ and $\rho_n \rightarrow \rho$ strongly in $L^2(D_L)$. If

$$\liminf_{n \rightarrow \infty} \tilde{\mathcal{E}}_{\varepsilon_n,L}(\rho_n) = +\infty,$$

there is nothing to prove. Otherwise, after passing to a subsequence, we may assume that the energies are uniformly bounded. Since $\delta > 0$, the gradient term together with the fixed mass gives a uniform $H^1(D_L)$ bound. Hence, after extracting a further subsequence,

$$\rho_n \rightharpoonup \rho \quad \text{weakly in } H^1(D_L).$$

Fix $\bar{\varepsilon} > 0$. For all sufficiently large n , $\varepsilon_n \leq \bar{\varepsilon}$. The ordering in (2.5) therefore gives

$$F_{\varepsilon_n}(s, q) \geq F_{\bar{\varepsilon}}(s, q).$$

For fixed $\bar{\varepsilon}$, the integrand $F_{\bar{\varepsilon}}$ is continuous and jointly convex in (s, q) . The associated integral functional is therefore weakly lower semicontinuous on $H^1(D_L)$. Together with the remaining terms, this yields

$$\liminf_{n \rightarrow \infty} \tilde{\mathcal{E}}_{\varepsilon_n,L}(\rho_n) \geq \tilde{\mathcal{E}}_{\bar{\varepsilon},L}(\rho).$$

Finally, let $\bar{\varepsilon} \downarrow 0$. Since $r_{\bar{\varepsilon}}(s) \downarrow s$, the nonnegative kinetic integrands increase pointwise to F_0 . The monotone convergence theorem then gives

$$\liminf_{n \rightarrow \infty} \tilde{\mathcal{E}}_{\varepsilon_n,L}(\rho_n) \geq \mathcal{E}_{0,L}(\rho).$$

For the recovery sequence, let $\rho \in \mathcal{A}_L$ satisfy $\mathcal{E}_{0,L}(\rho) < \infty$ and take the constant sequence $\rho_{\varepsilon} = \rho$. The same monotone convergence argument gives

$$\tilde{\mathcal{E}}_{\varepsilon,L}(\rho) \longrightarrow \mathcal{E}_{0,L}(\rho) \quad \text{as } \varepsilon \rightarrow 0^+.$$

Hence

$$\tilde{\mathcal{E}}_{\varepsilon,L} \xrightarrow{\Gamma} \mathcal{E}_{0,L} \quad \text{in strong } L^2(D_L).$$

It remains to restore the potential regularization. By (2.13),

$$\sup_{\rho \in \mathcal{A}_L} \left| \mathcal{E}_{\varepsilon,\sigma(\varepsilon),L}(\rho) - \tilde{\mathcal{E}}_{\varepsilon,L}(\rho) \right| \leq \sigma(\varepsilon) \int_{D_L} V dx \longrightarrow 0.$$

Thus the potential smoothing is a uniformly vanishing perturbation and does not change the Γ -limit.

The equicoercivity follows from the δ term and the fixed mass, which give a uniform $H^1(D_L)$ bound, together with the compact embedding $H^1(D_L) \hookrightarrow L^2(D_L)$. $\square$

The Γ -convergence result gives the corresponding convergence of minimum values and minimizers.

Corollary 2.6 (Convergence of fixed-box minimizers). *Under the assumptions of Theorem 2.5, let $\rho_{\varepsilon,L}$ and $\rho_{0,L}$ denote the minimizers of $\mathcal{E}_{\varepsilon,\sigma(\varepsilon),L}$ and $\mathcal{E}_{0,L}$ on $\mathcal{A}_L$, respectively. Then*

$$\min_{\rho \in \mathcal{A}_L} \mathcal{E}_{\varepsilon,\sigma(\varepsilon),L}(\rho) \longrightarrow \min_{\rho \in \mathcal{A}_L} \mathcal{E}_{0,L}(\rho), \quad \rho_{\varepsilon,L} \longrightarrow \rho_{0,L} \quad \text{strongly in } L^2(D_L). \quad (2.14)$$

Proof. Existence follows from the direct method on the fixed box, while strict convexity follows from the δ term as in Proposition 2.1. Hence both minimizers are unique.

The convergence of the minimum values and the subsequential convergence of $\rho_{\varepsilon,L}$ follow from Theorem 2.5 and equicoercivity. Every convergent subsequence has a minimizer of $\mathcal{E}_{0,L}$ as its limit. Since this minimizer is unique, all such limits coincide with $\rho_{0,L}$, and therefore the whole family converges. $\square$

3 Fourier pseudospectral discretization

In this section, we discretize the regularized density problem on a fixed periodic box by a Fourier pseudospectral method. We first define the discrete energy and its gradient, and then examine the Hessian structure that will be used in the local refinement. The positive mass-conservative projection required by the constrained optimization is discussed in the next subsection.

For clarity, we present the one-dimensional formulation on $D_L = [-L, L)$. The multidimensional discretization used later is obtained by the standard tensor-product extension. Let N be even, $x_j = -L + jh$, $j = 0, \dots, N-1$, and $h = 2L/N$. For $u, v \in \mathbb{R}^N$, define

$$\langle u, v \rangle_h = h \sum_{j=0}^{N-1} u_j v_j, \quad \|u\|_h^2 = \langle u, u \rangle_h. \quad (3.1)$$

3.1 Discrete energy, gradient, and Hessian structure

Let D_N denote the real Fourier pseudospectral first-derivative operator. For even N , we set the derivative of the Nyquist mode to zero, so that D_N maps real grid vectors to real grid vectors and satisfies

$$D_N^T = -D_N \quad (3.2)$$

with respect to the inner product (3.1). The action of D_N is evaluated by FFTs.

For a grid density ρ , let $q = D_N \rho$. The discrete regularized energy is

$$E_{\varepsilon,\sigma,N}(\rho) = h \sum_{j=0}^{N-1} \left[\frac{q_j^2}{8r_\varepsilon(\rho_j)} + V_j p_\sigma(\rho_j) + \frac{\beta}{2} \rho_j^2 + \frac{\delta}{2} q_j^2 \right], \quad (3.3)$$

subject to the discrete positivity and mass constraints

$$\mathcal{C}_N = \left\{ \rho \in \mathbb{R}^N : \rho_j \geq 0, \quad h \sum_{j=0}^{N-1} \rho_j = 1 \right\}. \quad (3.4)$$

The following proposition shows that the Fourier discretization retains the convex structure of the regularized density problem.

Proposition 3.1 (Discrete convexity). *Let r_ε be admissible in the sense of Definition 2.2. Let $\sigma > 0$ and assume $V_j \geq 0$, $\beta \geq 0$, and $\delta > 0$. Then $E_{\varepsilon,\sigma,N}$ is convex on $\mathcal{C}_N$ and admits a minimizer. If $\beta > 0$, it is strongly convex with respect to $\|\cdot\|_h$, and the minimizer is unique.*

Proof. By Lemma 2.3, $F_\varepsilon(s, q)$ is jointly convex. Since the map

$$\rho \mapsto (\rho_j, (D_N \rho)_j)$$

is linear for each j , the discrete kinetic contribution in (3.3) is convex in ρ . The potential contribution is convex because p_σ is convex and $V_j \geq 0$, while the β and δ terms are convex quadratics. Hence $E_{\varepsilon,\sigma,N}$ is convex on $\mathcal{C}_N$.

The feasible set $\mathcal{C}_N$ is compact and the discrete energy is continuous, so a minimizer exists. If $\beta > 0$, the term $\frac{\beta}{2} \|\rho\|_h^2$ gives strong convexity, and the minimizer is unique. $\square$

We next derive the gradient directly from the discrete energy. Let $q = D_N \rho$. With all non-linear operations understood pointwise, the gradient with respect to the inner product (3.1) is characterized by

$$\begin{aligned} G_{\varepsilon,\sigma,N}(\rho) &= \frac{1}{4} D_N^T \left(\frac{q}{r_\varepsilon(\rho)} \right) - \frac{1}{8} \frac{q^2 r'_\varepsilon(\rho)}{r_\varepsilon(\rho)^2} + V p'_\sigma(\rho) + \beta \rho + \delta D_N^T D_N \rho, \\ \left. \frac{d}{dt} E_{\varepsilon,\sigma,N}(\rho + t\eta) \right|_{t=0} &= \langle G_{\varepsilon,\sigma,N}(\rho), \eta \rangle_h. \end{aligned} \quad (3.5)$$

Thus the gradient used in the optimization is derived consistently from the same discrete energy.

We next record the second-order structure used in the local Newton refinement. At points where r_ε is twice differentiable, define

$$B = D_N - \text{diag} \left(\frac{q r'_\varepsilon(\rho)}{r_\varepsilon(\rho)} \right), \quad W = \text{diag} \left(\frac{1}{4r_\varepsilon(\rho)} \right). \quad (3.6)$$

The Hessian contribution of the regularized kinetic term admits the factorization

$$H_{\text{kin}} = B^T W B + \text{diag} \left(-\frac{q^2 r''_\varepsilon(\rho)}{8r_\varepsilon(\rho)^2} \right). \quad (3.7)$$

Since $r_\varepsilon > 0$ and $r''_\varepsilon \leq 0$ wherever the second derivative exists, both terms in (3.7) are positive semidefinite. The Hessian of the full discrete energy therefore has the decomposition

$$H = H_{\text{kin}} + \delta D_N^T D_N + \beta I + \text{diag}(V p''_\sigma(\rho)). \quad (3.8)$$

Hence H is symmetric positive semidefinite. If $\beta > 0$, it is positive definite. This structure motivates the Newton-PCG refinement introduced in Section 4.2.

For the shifted denominator $r_\varepsilon(s) = s + \varepsilon$, which is used in the Newton refinement below, $r'_\varepsilon = 1$ and $r''_\varepsilon = 0$. The diagonal correction in (3.7) then vanishes. The corresponding matrix-free Hessian action is recorded in Appendix B.

3.2 Positive mass-conservative projection

The constrained optimization requires a projection that preserves both nonnegativity and mass. We show that the corresponding spectral L^2 projection can be carried out exactly at the collocation points, reducing the constrained Fourier problem to a finite-dimensional simplex projection.

Let S_N denote the trigonometric interpolation space associated with the Fourier grid, and define

$$S_N^+ = \{v_N \in S_N : v_N(x_j) \geq 0, \quad j = 0, \dots, N-1\}. \quad (3.9)$$

Thus positivity is imposed at the collocation points rather than on the trigonometric interpolant over the whole interval.

For $z_N \in S_N$, consider the positive mass-conservative pseudospectral projection

$$\Pi_{N,h}^+ z_N = \underset{g_N \in S_N^+}{\text{argmin}} \frac{1}{2} \|g_N - z_N\|_{N,h}^2 \quad \text{subject to} \quad \int_{D_L} g_N \, dx = \int_{D_L} z_N \, dx, \quad (3.10)$$

where

$$\|v_N\|_{N,h}^2 = h \sum_{j=0}^{N-1} |v_N(x_j)|^2. \quad (3.11)$$

Let P_N denote the L^2 projection onto S_N .

Proposition 3.2 (Nodal form of the positive conservative projection). *Let $z_j = z_N(x_j)$. The pseudospectral projection (3.10) is equivalent to the nodal projection*

$$\operatorname{argmin}_{y \in \mathbb{R}^N} \frac{1}{2} \|y - z\|_h^2$$

subject to

$$y_j \geq 0, \quad h \sum_{j=0}^{N-1} y_j = \int_{D_L} z_N \, dx.$$

Proof. By the definition of the pseudospectral norm (3.11),

$$\|g_N - z_N\|_{N,h}^2 = h \sum_{j=0}^{N-1} |g_N(x_j) - z_N(x_j)|^2.$$

Moreover, the integral of a trigonometric polynomial is determined by its zero Fourier mode, so

$$\int_{D_L} g_N \, dx = h \sum_{j=0}^{N-1} g_N(x_j).$$

Since positivity is imposed at the collocation points, both the objective and the constraints reduce exactly to the stated nodal problem. $\square$

For the unit-mass constraint used below, the resulting nodal projection has the form

$$[P_{\mathcal{C}_N}(z)]_j = \max\{z_j - \lambda, 0\}, \quad h \sum_{j=0}^{N-1} [P_{\mathcal{C}_N}(z)]_j = 1, \quad (3.12)$$

where the scalar multiplier λ is uniquely determined by the mass constraint. Thus the positive mass-conservative projection reduces to a one-dimensional monotone equation for λ .

4 Potential-proximal optimization and KKT refinement

In this section, we solve the discrete convex minimization problem using a two-stage strategy. We first apply an accelerated proximal method to reduce the energy efficiently. We then refine the resulting state through the KKT system to obtain high-accuracy stationarity.

The discrete ground-state problem is

$$\min_{\rho \in \mathcal{C}_N} E_{\varepsilon, \sigma, N}(\rho). \quad (4.1)$$

Since the objective is convex and $\mathcal{C}_N$ is convex, every KKT point is a global minimizer.

To measure stationarity independently of the particular optimization method, for a fixed $\tau_r > 0$ define the projected-gradient residual

$$\mathcal{G}_{\tau_r}(\rho) = \frac{1}{\tau_r} [\rho - P_{\mathcal{C}_N}(\rho - \tau_r G_{\varepsilon, \sigma, N}(\rho))]. \quad (4.2)$$

For a feasible state ρ , the optimality condition for (4.1) is equivalent to $\mathcal{G}_{\tau_r}(\rho) = 0$. We therefore use $\|\mathcal{G}_{\tau_r}(\rho)\|_h$ as the final stationarity measure.

4.1 Potential-proximal FISTA

The potential regularization introduced in Section 2 has curvature $p''_\sigma(0) = 1/\sigma$, which becomes large as σ decreases. Including this contribution in the smooth part of FISTA would therefore lead to a large local curvature. To avoid this difficulty without changing the discrete energy, we use the splitting

$$E_{\varepsilon,\sigma,N}(\rho) = f_{\varepsilon,N}(\rho) + g_{\sigma,N}(\rho), \quad (4.3)$$

where

$$f_{\varepsilon,N}(\rho) = h \sum_j \left[\frac{(D_N \rho)_j^2}{8r_\varepsilon(\rho_j)} + \frac{\beta}{2} \rho_j^2 + \frac{\delta}{2} (D_N \rho)_j^2 \right] \quad (4.4)$$

contains the kinetic, interaction, and HOI-gradient contributions, while

$$g_{\sigma,N}(\rho) = h \sum_j V_j p_\sigma(\rho_j) + I_{\mathcal{C}_N}(\rho) \quad (4.5)$$

contains the regularized potential together with the positivity and mass constraints. Here $I_{\mathcal{C}_N}$ denotes the indicator function of $\mathcal{C}_N$.

The gradient of $f_{\varepsilon,N}$ with respect to (3.1) is

$$\nabla_h f_{\varepsilon,N}(\rho) = \frac{1}{4} D_N^T \left(\frac{D_N \rho}{r_\varepsilon(\rho)} \right) - \frac{1}{8} \frac{(D_N \rho)^2 r'_\varepsilon(\rho)}{r_\varepsilon(\rho)^2} + \beta \rho + \delta D_N^T D_N \rho, \quad (4.6)$$

where the nonlinear operations are understood pointwise.

Let y be the FISTA extrapolated state. To keep the smooth energy within its domain, the extrapolation is accepted only if $y \geq 0$. Otherwise, the momentum is restarted and y is reset to the current feasible iterate. For a step size $\tau > 0$, we then set

$$z = y - \tau \nabla_h f_{\varepsilon,N}(y).$$

The proximal step is

$$\rho^+ = \operatorname{argmin}_{\rho \in \mathcal{C}_N} \left\{ \frac{1}{2} \|\rho - z\|_h^2 + \tau h \sum_j V_j p_\sigma(\rho_j) \right\}. \quad (4.7)$$

The mass constraint introduces a single scalar multiplier. For a fixed multiplier λ , each positive component satisfies

$$s - z_j + \tau V_j p'_\sigma(s) + \lambda = 0, \quad s > 0. \quad (4.8)$$

The left-hand side is strictly increasing in s , since its derivative is $1 + \tau V_j p''_\sigma(s) > 0$. Hence the positive solution, when it exists, is unique. The j th component is zero when $z_j \leq \lambda$ and otherwise is given by the unique positive solution of (4.8). The multiplier λ is determined by the mass constraint

$$h \sum_j \rho_j^+ = 1.$$

Thus the proximal problem reduces to independent scalar nodal equations coupled only through the mass constraint.

For the unsmoothed linear potential, $p(r) = r$, the proximal step reduces to

$$\rho^+ = P_{\mathcal{C}_N}(z - \tau V),$$

so (4.7) can be viewed as a nonlinear extension of the positive mass-conservative projection in Proposition 3.2.

For the first-order stage, we use the Chambolle–Dossal variant of FISTA with backtracking on $f_{\varepsilon,N}$ [11, 13]. This stage is used to reduce the energy efficiently. Once the energy has stabilized and the projected residual is sufficiently small, the state is passed to the KKT refinement described below. The projected-gradient residual (4.2), rather than the energy change, is used as the final convergence criterion.

4.2 Newton–PCG refinement

We next refine the FISTA state by solving the KKT conditions of (4.1). They are

$$\begin{aligned} G_{\varepsilon,\sigma,N}(\rho) + \lambda \mathbf{1} - \mu &= 0, & h \mathbf{1}^T \rho &= 1, \\ \rho &\geq 0, & \mu &\geq 0, & \rho_j \mu_j &= 0. \end{aligned} \quad (4.9)$$

When the solution is interior, the inequality multiplier vanishes and (4.9) reduces to

$$G_{\varepsilon,\sigma,N}(\rho) + \lambda \mathbf{1} = 0, \quad h \mathbf{1}^T \rho = 1. \quad (4.10)$$

In the strongly convex regime $\beta > 0$, the Hessian is symmetric positive definite by (3.8). We therefore eliminate the mass multiplier rather than solve the indefinite KKT system directly. Let

$$r = G_{\varepsilon,\sigma,N}(\rho) + \lambda \mathbf{1}, \quad r_m = h \mathbf{1}^T \rho - 1$$

denote the stationarity and mass residuals. The Newton correction (d, d_λ) satisfies

$$Hd + d_\lambda \mathbf{1} = -r, \quad h \mathbf{1}^T d = -r_m.$$

Introducing the two symmetric positive definite solves

$$Hz = r, \quad Hw = \mathbf{1},$$

the Schur complement gives

$$d_\lambda = \frac{r_m - h \mathbf{1}^T z}{h \mathbf{1}^T w}, \quad d = -z - d_\lambda w. \quad (4.11)$$

Thus the interior Newton step requires only two linear systems with the same symmetric positive definite Hessian. A damped step is used to preserve positivity and globalize the iteration. If active constraints are encountered, we instead use a standard primal–dual active-set treatment [24].

Remark 4.1 (Preconditioning). To accelerate the PCG solves, we use a sparse variable-coefficient finite-difference approximation of the Fourier pseudospectral Hessian as a symmetric positive definite preconditioner. It retains the principal elliptic structure of the Hessian while allowing efficient sparse solves, and does not alter the discrete problem (4.1). Details are given in Appendix B.

5 Numerical experiments

In this section, we examine the spatial accuracy, regularization effects, and computational performance of the positive-conservative Fourier optimization (PCFO) method, and present representative two-dimensional examples.

Unless otherwise stated, we take $\beta = \delta = 10$, impose the unit-mass constraint, use the shifted denominator $r_\varepsilon(\rho) = \rho + \varepsilon$, and adopt the potential regularization (2.8).

Table 5.1: Fourier mesh refinement for $\varepsilon = \sigma = 10^{-2}$ using the harmonic trap with a C^∞ far-field periodic continuation. The comparison state is computed with $N_{\text{ref}} = 8192$.

N	$e_{\rho,2}$	rate	$e_{\text{res},\infty}$	rate	e_E	rate
32	2.519×10^{-2}	–	1.282×10^{-2}	–	5.831×10^{-3}	–
64	3.394×10^{-3}	2.89	1.422×10^{-3}	3.17	2.077×10^{-3}	1.49
128	1.190×10^{-4}	4.83	9.022×10^{-5}	3.98	5.597×10^{-6}	8.54
256	6.227×10^{-7}	7.58	2.526×10^{-7}	8.48	3.770×10^{-9}	10.54
512	6.415×10^{-11}	13.24	4.185×10^{-11}	12.56	4.174×10^{-14}	–

5.1 Spatial accuracy

We first examine the spatial accuracy of the Fourier discretization for a fixed regularized problem. To separate the spatial error from the regularization error, we work on the periodic interval $D_L = [-L, L)$ with $L = 32$ and fix $\varepsilon = \sigma = 10^{-2}$. For these parameters, the regularized density is smooth and its low-density transition is well resolved on the finer meshes.

The physical trapping potential is harmonic, $V_H(x) = x^2/2$. Its direct periodic extension is continuous at $\pm L$ but is not periodically smooth because $V'_H(-L) \neq V'_H(L)$. This lack of periodic smoothness eventually limits the accuracy of a Fourier discretization. To isolate the intrinsic spatial convergence of the method, we therefore modify the potential only near the boundary. With $R_0 = 0.75L$ and $R_1 = 0.90L$, we use

$$V_L^{\text{per}}(x) = (1 - \chi_L(x)) \frac{x^2}{2} + \chi_L(x) \frac{L^2}{2}, \quad (5.1)$$

where χ_L is an even C^∞ cutoff satisfying $\chi_L = 0$ for $|x| \leq R_0$ and $\chi_L = 1$ for $|x| \geq R_1$. Thus the harmonic trap is unchanged in the interior and becomes constant near the periodic boundary.

Let ρ_{ref} denote the solution on the finest comparison grid, and let $P_N \rho_{\text{ref}}$ be its orthogonal Fourier projection onto S_N . We measure the resolved-mode error, the unresolved reference tail, and the corresponding total state error by

$$\begin{aligned} e_{\text{res},2}(N) &= \|\rho_N - P_N \rho_{\text{ref}}\|_{h_N}, & e_{\text{res},\infty}(N) &= \|\rho_N - P_N \rho_{\text{ref}}\|_{\ell^\infty}, \\ e_{\text{tail},2}(N) &= \|(I - P_N) \rho_{\text{ref}}\|_{L^2(D_L)}, & e_{\rho,2}(N) &= (e_{\text{res},2}(N)^2 + e_{\text{tail},2}(N)^2)^{1/2}. \end{aligned} \quad (5.2)$$

The energy error is defined by

$$e_E(N) = |E_{\varepsilon,\sigma,N}(\rho_N) - E_{\varepsilon,\sigma,N_{\text{ref}}}(\rho_{\text{ref}})|. \quad (5.3)$$

Table 5.1 reports the errors relative to the $N_{\text{ref}} = 8192$ comparison state. The state and energy errors decrease rapidly, with increasing empirical orders, until the numerical floor is reached. This behavior provides clear numerical evidence of spectral-type spatial convergence for the fixed smooth regularized problem.

For comparison, we repeat the refinement using the unmodified harmonic potential $V_H = x^2/2$ on the periodic box. The two calculations are nearly indistinguishable on coarse and moderate meshes. At higher resolutions, however, the direct harmonic calculation develops an algebraic error floor, with the last resolved rates approaching second order. This behavior is consistent with the lack of periodic smoothness of V_H and is removed by the C^∞ far-field continuation. The contrast between the two cases is shown in Fig. 5.1.

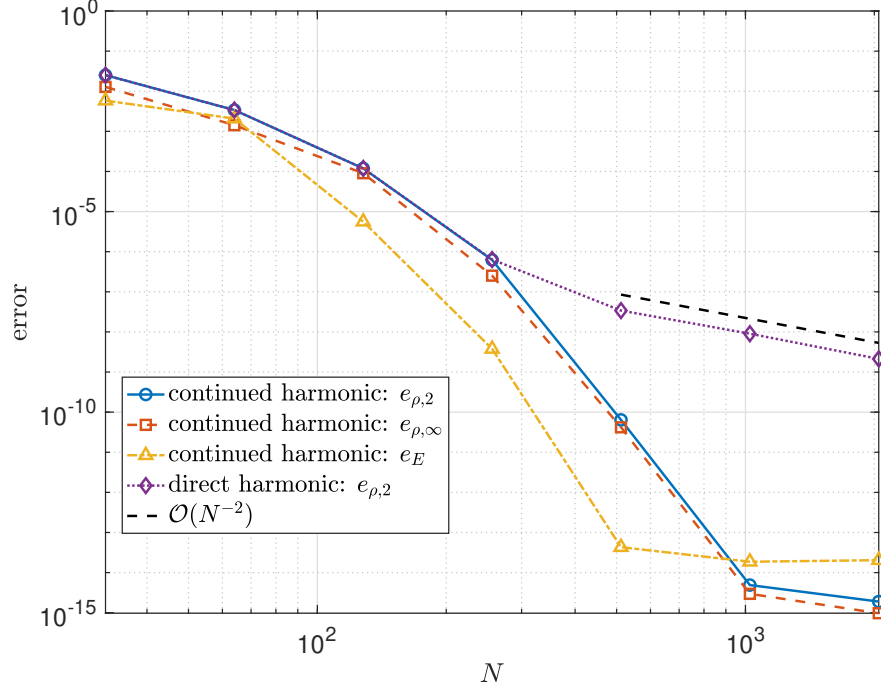


Figure 5.1: Fourier spatial convergence for the fixed regularized problem. The continued harmonic potential yields rapid decay of the state and energy errors, while the direct periodic treatment of the harmonic potential develops a high-resolution algebraic error floor.

5.2 Regularization effects

The regularized model contains two parameters associated with the kinetic and potential terms. We first examine the effect of the potential regularization parameter σ at fixed ε , including its influence on Fourier resolution. We then compare different regularizations of the density kinetic term as $\varepsilon \rightarrow 0$.

Potential regularization. We fix $\varepsilon = 10^{-2}$, $L = 32$, and $N = 4096$, and take $\sigma = \varepsilon^p$ with $p = 1, 2, 3, 4$. Let $\rho_{\varepsilon,\sigma}$ denote the minimizer with the regularized potential, and let $\rho_{\varepsilon,0}$ denote the minimizer obtained with the same kinetic regularization but the original linear potential. To quantify the perturbation introduced by p_σ , we define

$$\begin{aligned} B_E(\sigma) &= E_{\varepsilon,0}(\rho_{\varepsilon,\sigma}) - E_{\varepsilon,0}(\rho_{\varepsilon,0}), \\ B_\rho(\sigma) &= \|\rho_{\varepsilon,\sigma} - \rho_{\varepsilon,0}\|_{L^2(D_L)}. \end{aligned} \tag{5.4}$$

Here $E_{\varepsilon,0}$ denotes the energy with the same denominator r_ε and the original potential contribution $V\rho$.

Table 5.2 shows that both perturbations decrease approximately linearly with σ over the tested range. The minimum density exhibits the same scaling for small σ , indicating that the positive low-density background introduced by the potential regularization vanishes together with σ .

Reducing σ also makes the regularized low-density transition sharper. Indeed,

$$p'_\sigma(\rho) = \frac{\rho}{\sqrt{\rho^2 + \sigma^2}}$$

Table 5.2: Effect of the potential regularization scale at $\varepsilon = 10^{-2}$, $L = 32$, and $N = 4096$. The quantities B_E and B_ρ are defined in (5.4).

σ	$B_E(\sigma)$	$B_\rho(\sigma)$	$\min \rho$
10^{-2}	1.776×10^0	2.548×10^{-2}	6.449×10^{-5}
10^{-4}	1.849×10^{-2}	2.900×10^{-4}	6.740×10^{-7}
10^{-6}	1.849×10^{-4}	2.719×10^{-6}	6.743×10^{-9}
10^{-8}	1.849×10^{-6}	1.993×10^{-8}	6.744×10^{-11}

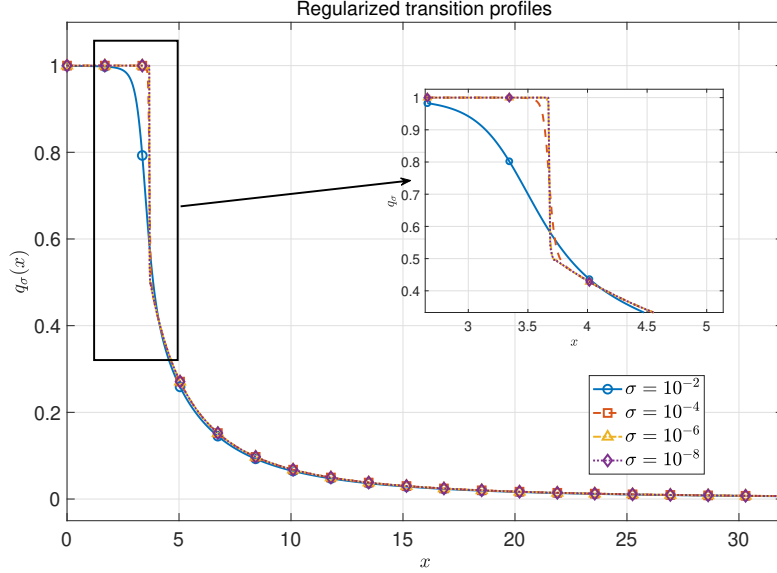


Figure 5.2: Transition profiles induced by the potential regularization. As σ decreases, the transition between the bulk and low-density regions becomes sharper. The inset enlarges the principal transition region. All computations use $\varepsilon = 10^{-2}$, $L = 32$, and $N = 4096$.

is close to one where $\rho \gg \sigma$ and approaches zero where $\rho \ll \sigma$. As σ decreases, the transition between these two regimes becomes increasingly sharp. This behavior is illustrated in Fig. 5.2.

To examine the resulting effect on Fourier resolution, we repeat the mesh refinement for $\sigma = 10^{-2}, 10^{-4}, 10^{-6}$, while keeping $\varepsilon = 10^{-2}$ and $L = 32$. We use the same C^∞ -continued harmonic potential as in Subsection 5.1, and measure the errors relative to an $N_{\text{ref}} = 65536$ comparison state. Figure 5.3 shows the resulting energy and state errors.

For $\sigma = 10^{-2}$, both errors rapidly enter the spectral regime and reach their numerical floors. As σ decreases, a longer pre-asymptotic range appears before the rapid spectral-type decay becomes visible. Thus a smaller σ reduces the potential regularization error but simultaneously sharpens the transition and requires a finer Fourier mesh. This bias-resolution tradeoff explains why the spatial-convergence experiment in Subsection 5.1 keeps the regularization parameters fixed rather than sending them to zero together with the mesh size.

Regularization of the kinetic term. We next isolate the effect of the kinetic regularization by fixing the potential regularization and the spatial discretization. We compare the shifted denominator (2.6) with the piecewise C^1 and C^2 denominators in (2.7). Throughout this test, we take

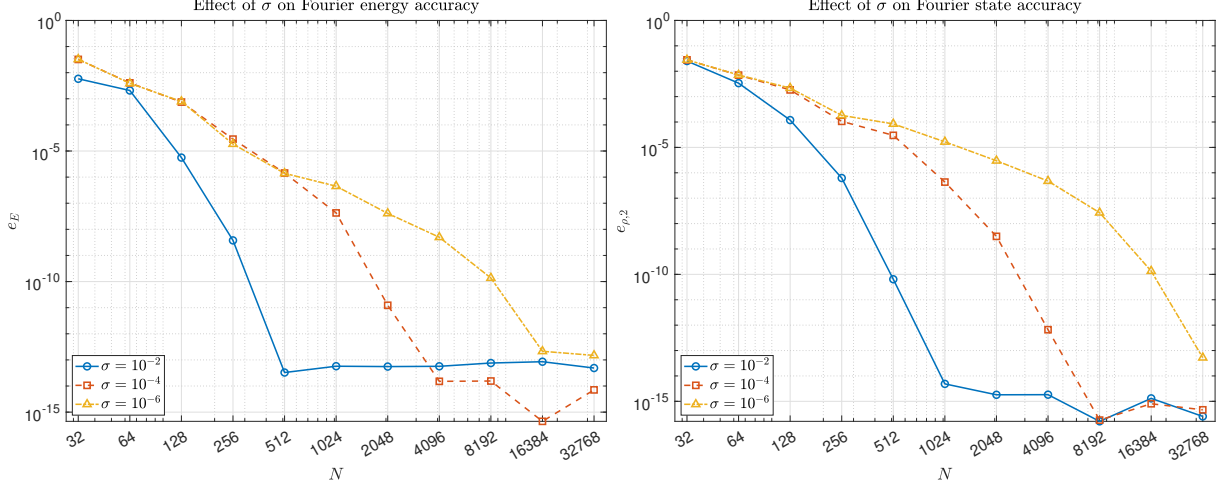


Figure 5.3: Effect of the potential regularization scale on Fourier accuracy. The left panel shows the energy error and the right panel shows the total spectral L^2 state error. Each curve refines a fixed- σ problem with $\varepsilon = 10^{-2}$, $L = 32$, and the harmonic-core C^∞ far-field continuation, relative to the $N_{\text{ref}} = 65536$ comparison state.

$\sigma = 10^{-2}$, $L = 32$, and $N = 512$, and use the same C^∞ -continued harmonic potential. The tested values are $\varepsilon = 10^{-1}, 5 \times 10^{-2}, 2 \times 10^{-2}, 10^{-2}, 5 \times 10^{-3}, 2 \times 10^{-3}, 10^{-3}$.

A common reference state ρ_{ref} is computed with the shifted denominator at $\varepsilon_{\text{ref}} = 10^{-5}$. To compare the three regularizations on the same basis, we measure the state error and evaluate the energy error with the common unregularized kinetic denominator $r_0(\rho) = \rho$:

$$\begin{aligned} e_\rho(\varepsilon) &= \|\rho_\varepsilon - \rho_{\text{ref}}\|_{L^2(D_L)}, \\ e_E(\varepsilon) &= |E_{0,\sigma}(\rho_\varepsilon) - E_{0,\sigma}(\rho_{\text{ref}})|. \end{aligned} \quad (5.5)$$

Here $E_{0,\sigma}$ denotes the energy with the original kinetic denominator $r_0(s) = s$ and the fixed potential regularization p_σ . Thus the energy comparison measures the effect of the computed states rather than the direct change of the regularized energy functional.

Figure 5.4 shows that the three regularizations have essentially the same asymptotic behavior as ε decreases. The common-functional energy error approaches second order whereas the state error is approximately first order. The piecewise C^1 denominator gives a modest reduction in error for the larger values of ε , but neither piecewise regularization shows a clear asymptotic rate advantage over the shifted denominator.

5.3 Solver performance

We next examine the computational performance of the two-stage optimization strategy. Throughout this subsection, we use the shifted denominator $r_\varepsilon(\rho) = \rho + \varepsilon$. We first illustrate the respective roles of FISTA and the Newton-PCG refinement, and then examine how the two regularization scales affect the computational cost.

FISTA and Newton refinement. We consider the representative case $\varepsilon = 10^{-2}$, $\sigma = 10^{-4}$, $L = 32$, and $N = 4096$. A common FISTA iteration is first run until the switching criterion is met. From the resulting state, one branch continues with FISTA, while the other proceeds with the

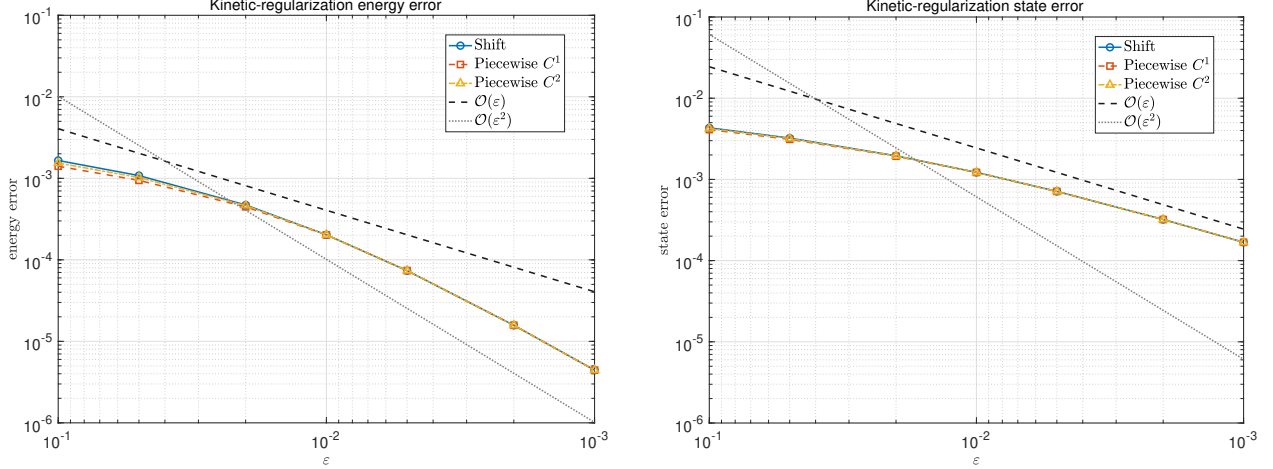


Figure 5.4: Effect of the kinetic regularization. The shifted and piecewise C^1/C^2 denominators are compared using the same potential regularization, Fourier resolution, and reference state. The left panel shows the L^2 state error and the right panel shows the energy error evaluated with the common unregularized kinetic denominator $r_0(s) = s$. The dashed and dotted lines indicate first- and second-order slopes, respectively.

Newton–PCG refinement. The two strategies therefore share exactly the same first-order history up to the switch.

Since FISTA and Newton steps have different computational costs, Fig. 5.5 reports the convergence histories against cumulative wall-clock time. The common prefix uses a single time history, after which the incremental time of each branch is added to the common switch time. The left panel shows the energy error relative to the final two-stage solution, while the right panel shows the projected-gradient residual. The marker denotes the switch to the Newton–PCG refinement.

At the switch, after 3025 FISTA iterations and 106.31 s, the energy has already reached its numerical plateau, whereas the projected-gradient residual remains appreciably larger than the final tolerance. Continued FISTA reduces this residual only slowly, while the Newton–PCG refinement rapidly reaches high-accuracy stationarity. In the shared-prefix experiment, the two-stage method reaches a projected-gradient residual of 3.11×10^{-11} at a cumulative time of 107.02 s. Continued FISTA has a residual of 3.25×10^{-8} after 50000 iterations and 1790.63 s. The comparison therefore highlights the complementary roles of the two stages: FISTA provides the main energy minimization, while Newton–PCG efficiently resolves the remaining stationarity error.

Dependence on the regularization scales. We next examine how the computational cost changes as the regularization parameters decrease. To isolate the two effects, we first vary ε at fixed $\sigma = 10^{-2}$ and then vary σ at fixed $\varepsilon = 10^{-2}$. In all cases, $r_\varepsilon(\rho) = \rho + \varepsilon$.

All runs use the same Fourier resolution $N = 4096$, initial density, and solver parameters. Thus the comparison reflects changes in the conditioning of the regularized problem rather than changes in the number of Fourier degrees of freedom. The results are summarized in Table 5.3. Here wall-clock time comprises the production FISTA stage and the Newton–PCG KKT refinement, including their standard convergence diagnostics, but excludes problem construction, initialization, file I/O, plotting, and experiment-specific postprocessing.

At fixed $\sigma = 10^{-2}$, decreasing ε makes the regularized kinetic term increasingly stiff in low-density regions. This is reflected in the larger FISTA iteration counts and total solution times in

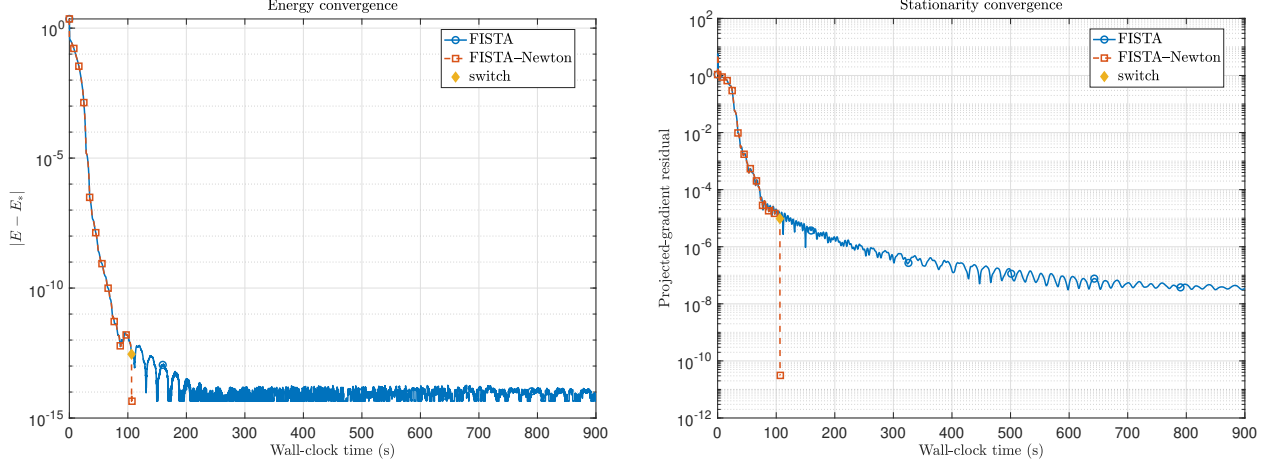


Figure 5.5: Convergence histories for continued FISTA and the two-stage FISTA–Newton strategy with $\varepsilon = 10^{-2}$, $\sigma = 10^{-4}$, $L = 32$, and $N = 4096$. The left panel shows the energy error relative to the final two-stage solution, and the right panel shows the projected-gradient residual. The two curves share exactly the same FISTA history up to the marker, which denotes the switch to the Newton–PCG refinement; thereafter one branch continues FISTA and the other uses Newton–PCG.

Table 5.3. The smallest ε case reaches the prescribed FISTA iteration limit and has a somewhat larger final residual than the other runs.

By contrast, decreasing σ at fixed $\varepsilon = 10^{-2}$ does not produce the same monotone increase in total cost over the tested range. Although a smaller σ sharpens the potential transition and changes the cost of the proximal and Newton stages, the potential-proximal splitting keeps the large curvature $p''_\sigma(0) = 1/\sigma$ outside the smooth FISTA gradient. This behavior is consistent with the motivation for the splitting in Subsection 4.1.

5.4 Two-dimensional examples

We conclude the numerical study with two two-dimensional examples to illustrate the applicability of the method to more general trapping potentials. The effects of the Fourier resolution, regularization parameters, and optimization accuracy have already been examined in the preceding subsections, so no additional convergence study is performed here.

Throughout this subsection, we use the shifted denominator $r_\varepsilon(\rho) = \rho + \varepsilon$, the potential regularization p_σ , and the parameters $\beta = \delta = 10$, $\varepsilon = 10^{-2}$, and $\sigma = 10^{-4}$. The computations are carried out on the periodic box $[-16, 16]^2$ with $N_x = N_y = 512$ Fourier collocation points. As in the one-dimensional experiments, the harmonic part of the potential is left unchanged in the central region and smoothly continued to a constant near the periodic boundary.

We first consider the anisotropic harmonic trap

$$V_{\text{ah}}(x, y) = \frac{1}{2} (\gamma_x^2 x^2 + \gamma_y^2 y^2), \quad \gamma_x = 1, \quad \gamma_y = 2. \quad (5.6)$$

The resulting ground-state density is shown in the left panel of Fig. 5.6. The stronger confinement in the y -direction produces the expected anisotropic density profile.

As a second example, we add a square optical lattice to the same harmonic confinement,

$$V_{\text{ol}}(x, y) = V_{\text{ah}}(x, y) + V_0 [\sin^2(kx) + \sin^2(ky)], \quad (5.7)$$

Table 5.3: Computational cost of the two-stage solver for different regularization scales. The shifted denominator $r_\varepsilon(\rho) = \rho + \varepsilon$ is used in all cases, with the same initialization, Fourier resolution $N = 4096$, and solver parameters. “FISTA” and “Newton” denote the iteration counts in the two stages, and “PCG” is the accumulated number of inner PCG iterations. The last column reports the final projected-gradient residual $\|\mathcal{G}_{\tau_r}(\rho)\|_h$ (4.2). The asterisk indicates that the FISTA stage reached the prescribed iteration limit.

ε	σ	FISTA	Newton	PCG	time (s)	$\ \mathcal{G}_{\tau_r}(\rho)\ _h$
10^{-2}	10^{-2}	3422	8	125	93.50	5.04×10^{-11}
10^{-3}	10^{-2}	9042	10	187	201.64	2.54×10^{-10}
10^{-4}	10^{-2}	20000*	14	280	519.77	1.15×10^{-9}
10^{-2}	10^{-3}	2992	10	186	99.28	3.93×10^{-11}
10^{-2}	10^{-4}	3025	18	341	107.02	3.11×10^{-11}

with $V_0 = 5$ and $k = \pi/2$. The additional periodic structure creates multiple local wells and leads to the more structured density profile shown in the right panel of Fig. 5.6. This example illustrates that the potential-proximal formulation is not restricted to a single-well trapping potential.

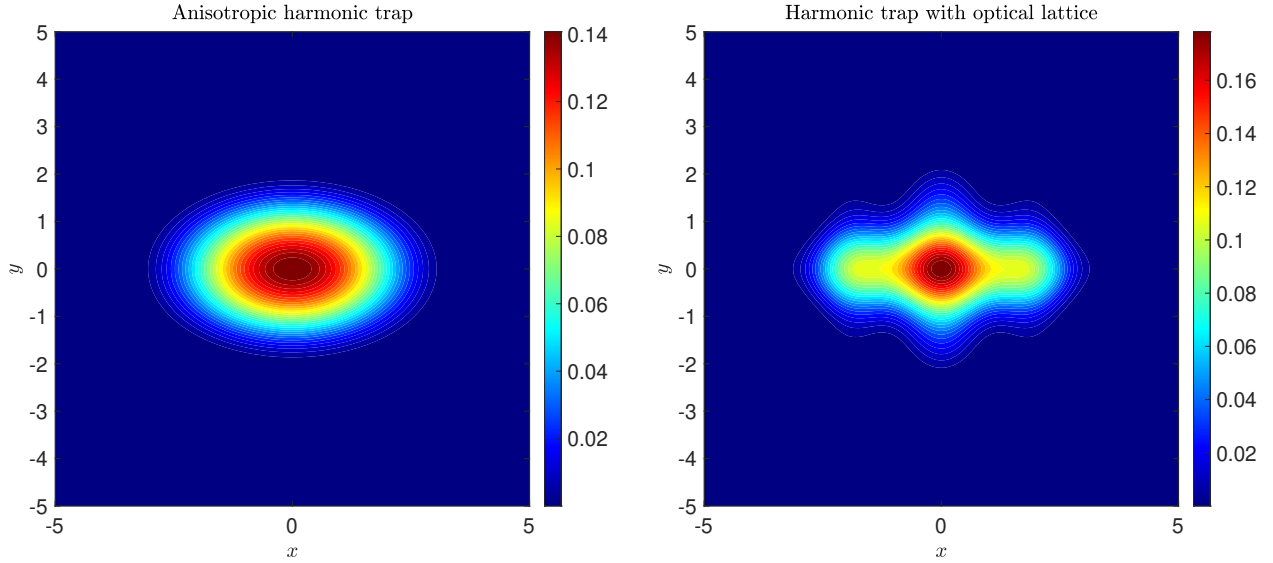


Figure 5.6: Two-dimensional ground-state densities computed on $[-16, 16]^2$ with $N_x = N_y = 512$, with the central region shown. The left panel corresponds to the anisotropic harmonic trap (5.6), while the right panel includes the square optical lattice (5.7). Both computations use $\varepsilon = 10^{-2}$, $\sigma = 10^{-4}$, and $\beta = \delta = 10$.

6 Conclusion

We have developed a convex Fourier pseudospectral method for computing ground states of Bose–Einstein condensates with higher-order interactions in the density formulation. The singular kinetic term and the vacuum behavior of the potential are treated by two compatible convex regularizations, whose fixed-box energies Γ -converge to the original density energy as the regularization parameters

vanish.

The resulting discretization preserves mass and nodal positivity. A potential-proximal FISTA iteration provides the main minimization, followed by a Newton–PCG refinement for high-accuracy stationarity. Numerical experiments demonstrate spectral-type convergence for fixed smooth regularized problems and reveal the bias–resolution tradeoff associated with the potential regularization. A joint analysis of the box, Fourier, and regularization limits remains for future work.

A KKT system for the positive conservative projection

For completeness, consider the nodal projection

$$\min_y \frac{1}{2} \|y - z\|_h^2 \quad \text{subject to} \quad y \geq 0, \quad h\mathbf{1}^T y = 1.$$

Its KKT conditions are

$$y - z - \lambda\mathbf{1} - \mu = 0, \quad y \geq 0, \quad \mu \geq 0, \quad y_j \mu_j = 0.$$

It follows that

$$y_j = \max\{z_j + \lambda, 0\},$$

where λ is determined by the monotone mass equation

$$h \sum_j \max\{z_j + \lambda, 0\} = 1.$$

This gives the scalar characterization of the positive mass-conservative projection used in Section 3.

B Newton–PCG implementation

We record the Hessian action and finite-difference preconditioner used in the Newton–PCG refinement. For compactness, let

$$a = r_\varepsilon(\rho), \quad a' = r'_\varepsilon(\rho), \quad a'' = r''_\varepsilon(\rho), \quad q = D_N \rho, \quad \dot{q} = D_N \eta.$$

At points where r_ε is twice differentiable, the Hessian action of the regularized kinetic term is

$$H_{\text{kin}}(\rho)[\eta] = \frac{1}{4} D_N^T \left[\frac{\dot{q}}{a} - \frac{q a'}{a^2} \eta \right] - \frac{1}{8} \left[2q \dot{q} \frac{a'}{a^2} + q^2 \left(\frac{a''}{a^2} - \frac{2(a')^2}{a^3} \right) \eta \right]. \quad (\text{B.1})$$

The full Hessian action is

$$H(\rho)[\eta] = H_{\text{kin}}(\rho)[\eta] + V p''_\sigma(\rho) \eta + \beta \eta + \delta D_N^T D_N \eta. \quad (\text{B.2})$$

For the PCG preconditioner, let D_{FD} denote the periodic forward-difference operator and define

$$B_{\text{FD}} = D_{\text{FD}} - \text{diag} \left(\frac{q a'}{a} \right), \quad W = \text{diag} \left(\frac{1}{4a} \right), \quad R_{\text{kin}} = \text{diag} \left(-\frac{q^2 a''}{8a^2} \right).$$

We use the sparse approximation

$$P_{\text{FD}} = B_{\text{FD}}^T W B_{\text{FD}} + R_{\text{kin}} + \delta D_{\text{FD}}^T D_{\text{FD}} + \text{diag}(\beta + V p''_\sigma(\rho)). \quad (\text{B.3})$$

In the strongly convex regime, P_{FD} is symmetric positive definite. It is used only as a preconditioner; the discrete energy, gradient, and Hessian action remain Fourier pseudospectral.

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